Ronaldo Franco Jaldin

Machine Learning Engineer

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Machine learning engineer with experience in code retrieval, software repair, and computer vision.

Education

HSE University — Moscow Graduated 2026

BSc, Applied Mathematics and Informatics

Experience

Freelance Machine Learning Engineer Apr 2025–present

- Delivered a camera-based attendance system with face detection and anti-spoofing checks.
- Developed video analytics to recognize coffee-shop products, count production, and generate timestamped reports; prepared training data with model-assisted annotation.
- Work with Docker and Kubernetes in development workflows.
- Use AI coding tools (Claude, Codex) to speed up development and testing.

Research Assistant — Huawei Research Institute Mar 2024–Mar 2025

- Built reproducible test-environment pipelines across repositories and contributed Docker environments to the SWE-bench harness.
- Built a patch/file retrieval pipeline and fine-tuned Jina embeddings with QLoRA, reaching 0.62 MRR on SWE-bench Verified above the off-the-shelf baseline.
- Investigated how Llama 3.1 prompting and repeated sampling affect issue resolution with oracle-retrieved context; experiments resolved 77 % of evaluated issues.
- Co-authored an arXiv preprint on test-feedback-driven program repair (2024).

Student Researcher — HSE Computational Biology Lab Mar–Sep 2023

Worked in C++ on parallelization and concurrency optimizations for the Crux proteomics toolkit.

Technical Service

IOI 2025 Scientific Committee — Problem setting, testing, and contest design.

BSc Thesis — GFlowNet Post-training for Molecule Generation

Studied GFlowNet post-training of BioT5+ to generate multiple candidate molecules from text. Adding supervised sequences from ground-truth examples made early training more stable and yielded slightly more distinct, valid molecules similar to those examples.

Honors and Competitions

- **ICPC World Finalist** (2022) — **Honorable Mention**, Bangladesh.
- ICPC South America Regional — **3rd** (2020), 10th (2019), and 15th (2018).
- Bolivia Preliminary Contest — **1st place** (2018, 2019, 2020, and 2025).
- IEEEXtreme global programming competition — **11th, 22nd, 19th, and 38th worldwide** (2018–2021).
- Topcoder Open South America Regional — **2nd place** (2019).
- International Olympiad in Informatics (IOI) — participant (2014, 2015, and 2017).
- Ibero-American Olympiad in Informatics — **bronze** (2014, 2015); Bolivian Olympiad in Informatics — **gold** (2016), **silver** (2013, 2014, 2015).

Technical Skills

Programming	Python · C++ · SQL · Bash
Software	FastAPI · PostgreSQL · Docker · Kubernetes · Git/GitHub · GitHub Actions
ML / Search	PyTorch · Transformers · LoRA/QLoRA · FAISS
Vision	OpenCV · YOLO · ByteTrack · ONNX Runtime

Languages: Spanish (native) · English (fluent) · Russian (intermediate)

Teaching and Coaching. Taught algorithms at UMSS (2025), introductory machine learning at UMSS (2026), and private online LLM classes (2025). Coached school students and co-coached ICPC teams in Bolivia (2016–2023); taught at UMSS ICPC camps (2018–2019).